

CSC/SDS 109: Communicating with Data

A4: Digital Visualization—CEEDS Carbon Emissions Data

Goals:

- Find interesting trends or patterns in the CEEDS Carbon Emissions Data
- Compose an interactive, digital visualization of the emissions data

Important Notes:

- Notice that part 1 of this assignment spans 2 weeks. The quality of what you produce should reflect two full weeks of work; do your best to deliver visualization(s) that are creative, precise, and meaningful.
- You may work alone or in small groups (2-3). If you choose to work in a group, the quantity and quality of your visualization(s) should scale accordingly.
- This class covers Tableau. If you already know a different way to make interactive, digital visualizations you are welcome to use that method. However, TAs and teaching staff may not be able to help you as easily if you get stuck.

Instructions – Part 1 DUE IN CLASS ON VIEWING DAY

For this assignment, you will analyze data from CEEDS on carbon emissions from the Sustainability Indicator Management and Analysis Platform (SIMAP), which is hosted by the University of New Hampshire. You can find links to the relevant datasets on the course website under “Assignments”:

- **peer_carbon_comparisons.xlsx**: contains carbon emissions across scopes for Smith and peer institutions (scopes tab), and carbon emissions across categories for Smith and peer institutions (categories tab) across years. These data are self-reported by institutions. Information on scopes and categories is available here: <https://unhsimap.org/cmap/resources/EFVersions>
- **Smith_Carbon_Report_2020-2026.csv**: contains detailed Smith emissions data from 2020-2026. (Remember, 2020, 2021, and possibly 2022 may look odd due to COVID.)

Your task for this assignment is to create an interactive dashboard that enables a user to learn something about Smith's emissions. Your dashboard must include at least 2 different interactions. It must also have a clear purpose. That purpose might be to communicate a specific analytical insight, to motivate action from the reader, to teach the reader about emissions, etc.; you choose.

Start by exploring the available datasets. What does each capture? What questions do you have about the data? What questions could the data help you answer? Browsing the SIMAP website (<https://unhsimap.org/>) may help you understand the data. You might also want to stop by CEEDS and see if anyone there wants to talk about Smith's carbon emissions / sustainability efforts. (They might even have ideas for visuals that would be meaningful or helpful to them!)

Competency Demonstration – Part 2, start in class on viewing day

On viewing day (see course calendar), bring your complete visualization(s) to class. Viewing day will be reserved to engage with each other's visualizations and provide feedback. You will also get verbal feedback from the instructor on viewing day. One way to provide feedback and demonstrate competencies is to write a thoughtful and actionable critique of someone else's visualization(s). If you choose to do this, be sure to cite visualization science and theories to support your suggested changes and compliments.

This is a competency-based course. After you complete your assignment, write a short, succinct, and clear reflection explaining which core competencies you demonstrated in your work and how.

It is recommended that you use this assignment to demonstrate 2-3 of the following competencies:

- Map data to appropriate visual channels
- Apply perceptual concepts to visualization design
- Accurately encode data values via color
- Appropriately use visual embellishment
- Create a data story using storytelling techniques
- Write a thoughtful and actionable critique of someone else's work
- Redesign a visualization based on critiques
- Positively contribute to the classroom community

If you choose to demonstrate “Write a thoughtful and actionable critique of someone else's work”, include your critique and an image of the visualization it critiques in your submission.

If you choose to demonstrate “Redesign a visualization based on critiques”, include the critiques you received and a clear, concise, detailed sketch/outline of how you would revise your visualization based on the critiques.

Submission

Create a PDF with images of your visualization(s) and your write up from part 2. Submit your completed work as a PDF on Gradescope.